PRAKHAR

Senior Software Engineer | Full Stack | Node.js | React.js | PostgreSQL
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Summary

Senior Software Engineer with **4+ years of experience** building enterprise SaaS, ERP, and Industrial IoT applications. Strong in Node.js, REST APIs, PostgreSQL, query optimization, and system integration. Delivered **12+ ERP modules** and **300+ APIs**, with hands-on experience in backend architecture, production systems, and end-to-end engineering delivery. Progressed from **Software Development Engineer** to **Software Manager** while remaining actively involved in development.

Experience

Software Manager Oct 2024 – Oct 2026
YM Automation Private Limited Coimbatore, Tamil Nadu

- Architected and delivered enterprise SaaS, ERP, and Industrial IoT applications using Node.js, React, PostgreSQL, and industrial communication protocols, supporting enterprise workflows and real-time operations.
- Owned end-to-end delivery of **7–8 projects** across approximately **10 customer organizations**, driving requirements, architecture, estimation, technical planning, deployment, and production delivery.
- Led industrial IoT deployments integrating PLCs and up to **65 machines** across multiple sites, enabling monitoring of machine status, production, energy, and operational events.
- Developed reusable frontend components and backend **MQTT/Modbus** integration setups, promoting consistent implementation across projects and reducing duplicated engineering effort.
- Led engineering execution across backend, frontend, mobile, and UI/UX, combining hands-on architecture, development, and debugging with code reviews, mentoring, and delivery coordination.

Software Development Engineer Jul 2022 – Oct 2024
YM Automation Private Limited Remote, Bengaluru, Karnataka

- Independently designed, developed, and deployed **12+ production software projects**, owning end-to-end full-stack implementation and delivery.
- Built **300+ REST APIs** and a PostgreSQL data model with **100+ tables** for an enterprise ERP platform, implementing business logic, authentication, RBAC, and transaction-based workflows.
- Optimized PostgreSQL queries and backend data retrieval for Manufacturing Process Tracker BOM reporting, implementing cursor-based pagination and dynamic filtering to support reports containing **1,000+ rows per project**.
- Designed PostgreSQL schemas and optimized SQL queries using indexing, transactions, relationships, and JSONB to support production application workloads.
- Developed industrial data acquisition and integration solutions using MQTT, Modbus TCP/RTU, TCP sockets, and PLC communication, supporting machine monitoring and data processing.

Projects

Enterprise ERP & SaaS Platform (React.js | Node.js | Express.js | PostgreSQL)

- Designed and developed a **12+ module** ERP platform covering HRMS, payroll, inventory, CRM, project management, and enterprise business workflows.
- Built **300+ REST APIs** and a PostgreSQL data model with **100+ tables**, implementing business logic, authentication, RBAC, and transaction-based workflows.
- Delivered a multi-company SaaS platform using separate PostgreSQL databases per company, currently supporting **5 onboarded companies** (including YM Automation) and approximately **~300–400 employees** represented in the system.

- **Clients:** Cogent Automation Pvt. Ltd.; Medicare Enviro System; MWP Energy Infra Private Limited; Monsoon Ex Brand.

Production Tracking & Workflow System (React.js | Node.js | Express.js | PostgreSQL)

- Engineered a production tracking system supporting **12-stage workflows** with configurable routing and branching logic for complex production operations.
- Optimized BOM report retrieval through dynamic filtering and cursor-based pagination, supporting **1,000+ rows per project** while addressing backend query performance bottlenecks.
- Implemented APIs and business logic for stage transitions, production status, quantity tracking, and workflow visibility across production operations.
- **Client:** Cogent Automation Pvt. Ltd.

Industrial IoT Monitoring Platform (React.js | Node.js | PostgreSQL | MQTT | Modbus | TCP/IP)

- Developed industrial monitoring applications integrating PLCs and up to **65 machines** to capture production, energy, machine status, alarms, and operational events.
- Implemented industrial data acquisition and communication using **MQTT, Modbus TCP/RTU, and TCP sockets** to process and persist machine data.
- Built monitoring dashboards, historical analytics, and reporting workflows to provide operational visibility into connected industrial systems.
- **Clients:** Salzer Electronics Limited – Unit II; Armor Grandeur Pvt. Ltd.; Medicare Enviro System; Cogent Automation Pvt. Ltd.; TE Connectivity India Pvt. Ltd. *(via Cogent Automation Pvt. Ltd.)*; The Tata Power Company Limited *(via Renso Engineering Pvt. Ltd.)*.

Skills

Frontend
React.js, HTML5, CSS3, Axios, State Management, Responsive UI, Javascript

Backend
Node.js, Express.js, JavaScript, REST APIs, WebSockets, Authentication, RBAC, API Development, Business Logic

Database
PostgreSQL, SQL, Database Design, Query Optimization, Indexing, Transactions, JSONB, MongoDB

Cloud & Deployment
Microsoft Azure, AWS EC2, Linux, Nginx, PM2, GitHub Actions, SSL, DNS

Industrial IoT
MQTT, Modbus TCP/RTU, TCP/IP, TCP Sockets, PLC Integration, Real-time Data Processing

Engineering
Full-Stack Development, System Architecture, Database Architecture, API Development, Performance Optimization, Authentication, RBAC, Code Review, Debugging

Education

Computer Science and Engineering — Tumakuru, Karnataka
Siddaganga Institute of Technology, Tumakuru — Bachelor of Engineering